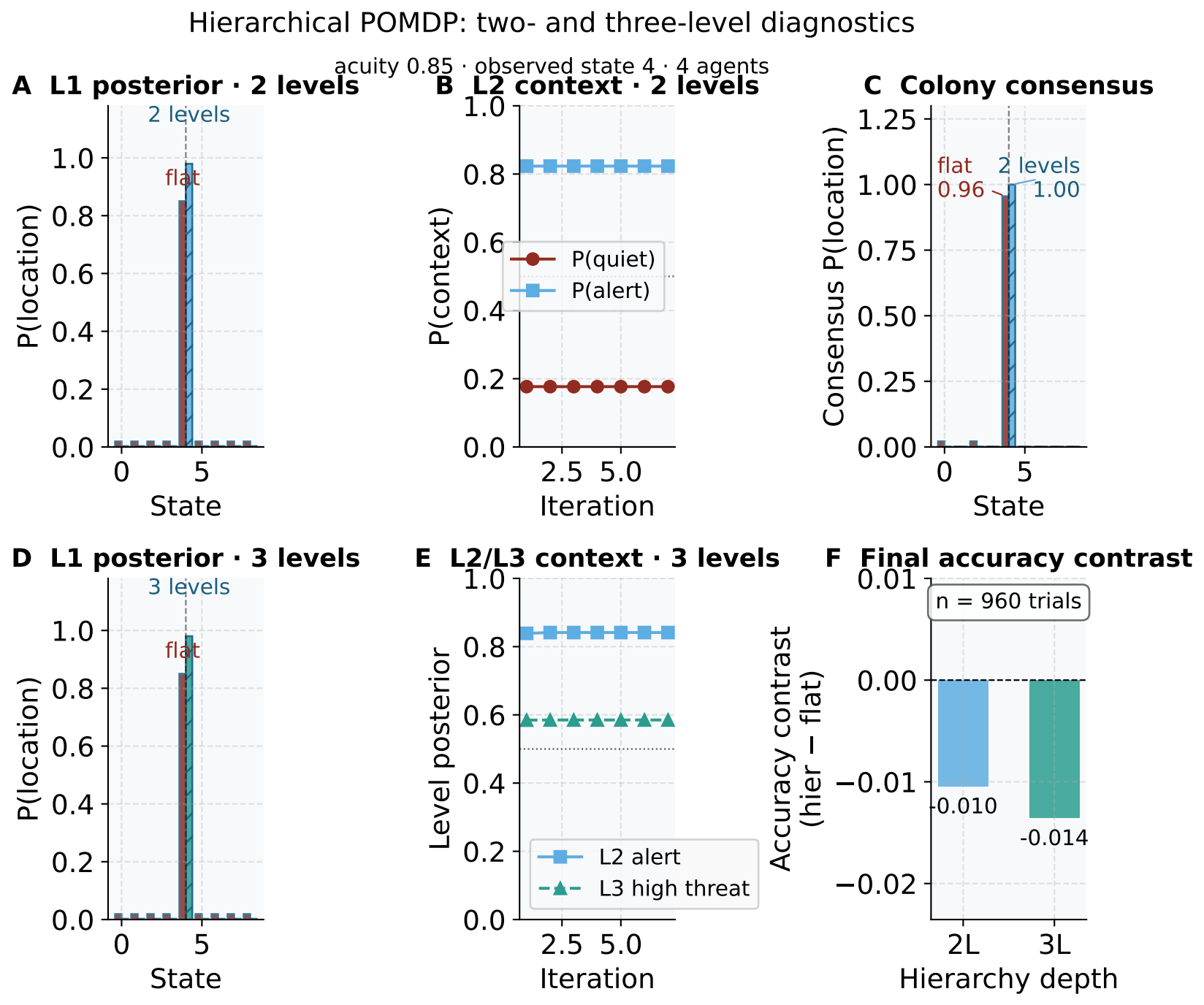
Hierarchical POMDP: two- and three-level diagnostics
acuity 0.85 · observed state 4 · 4 agents
A L1 posterior · 2 levels
2 levels
flat
P(location)
State
B L2 context · 2 levels
P(quiet)
P(alert)
P(context)
Iteration
C Colony consensus
flat
0.96
2 levels
1.00
Consensus P(location)
State
D L1 posterior · 3 levels
3 levels
flat
P(location)
State
E L2/L3 context · 3 levels
L2 alert
L3 high threat
Level posterior
Iteration
F Final accuracy contrast
n = 960 trials
-0.010
-0.014
Accuracy contrast (hier − flat)
2L
3L
Hierarchy depth